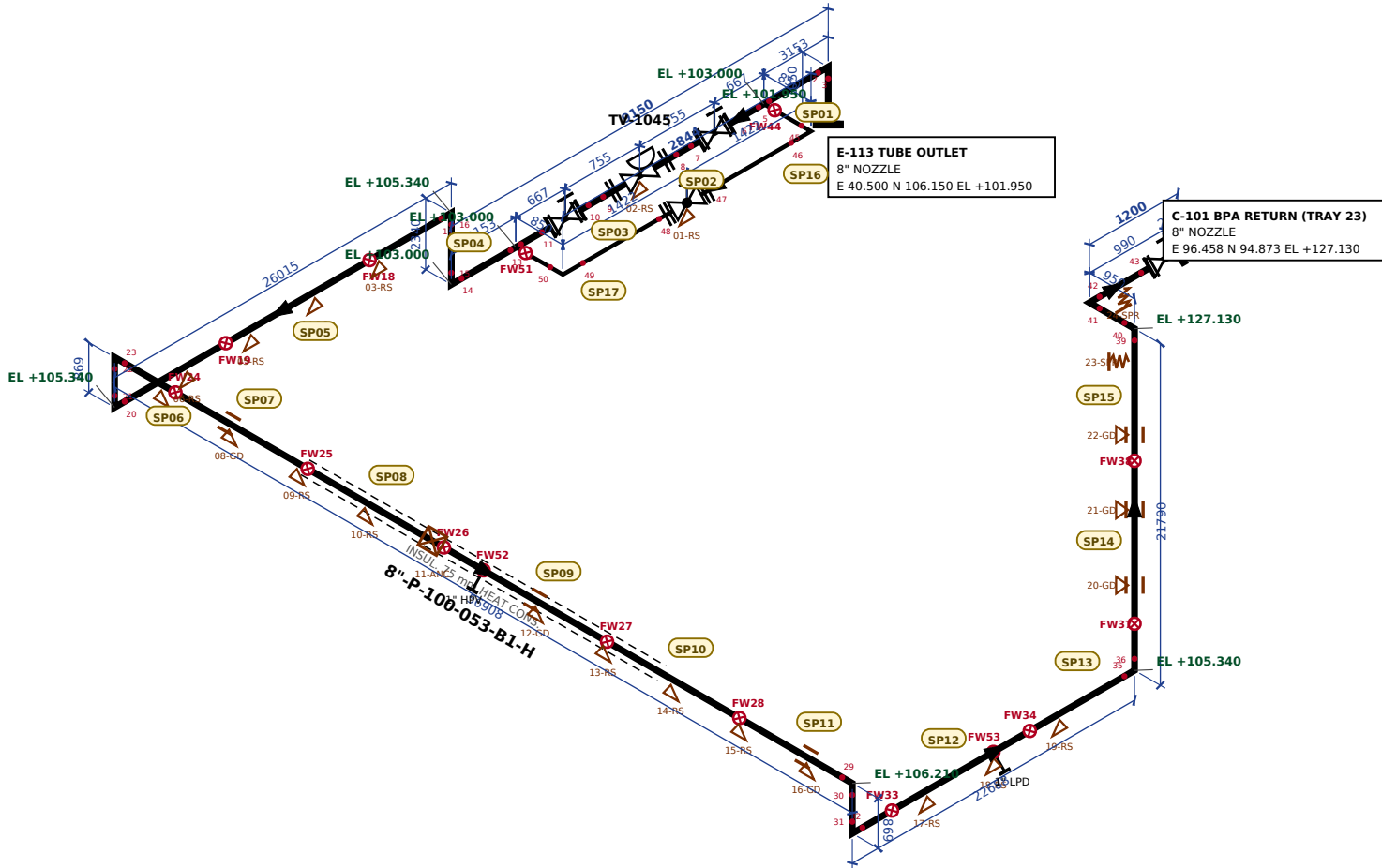
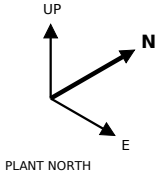


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	8"	8
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	8"	134.3 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	8"	10
4	TEE RED., BW, SCH 40, ASTM A234 WPB	8" x 6"	2
5	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	6"	2.6 m
6	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	6"	2
7	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	6"	2
8	SOCKOLET 3000# (branch), ASTM A105	1"	2
9	ERECTION: GATE VALVE OS&Y CL300 RF, A216 WCB, trim 8	8"	3
10	CONTROL VALVE TV-1045 CL300 RF (by I&C)	8"	1
11	GLOBE VALVE CL300 RF, A216 WCB, trim 8	6"	1
12	GASKET SPW 304/graphite, outer ring, CL300 B16.20	8"	9
13	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	8"	9
14	GASKET SPW 304/graphite, outer ring, CL300 B16.20	6"	2
15	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	6"	2
16	PIPE SUPPORTS (std. dwgs): anchor x1, guide x6, rest x15, spring x2	-	24

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	8"	635	SP01-2	8"	2670	SP01-3	8"	170
SP02-1	8"	20	SP03-1	8"	20	SP04-1	8"	170
SP04-2	8"	2670	SP04-3	8"	1731	SP04-4	8"	5650
SP05-1	8"	11700	SP06-1	8"	8055	SP06-2	8"	260
SP06-3	8"	2450	SP07-1	8"	11700	SP08-1	8"	11700
SP09-1	8"	11700	SP10-1	8"	11700	SP11-1	8"	7048
SP11-2	8"	260	SP11-3	8"	2450	SP12-1	8"	11700
SP13-1	8"	7928	SP13-2	8"	2450	SP14-1	8"	11700
SP15-1	8"	7030	SP15-2	8"	340	SP15-3	8"	366
SP16-1	6"	443	SP16-2	6"	874	SP17-1	6"	874
SP17-2	6"	443						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=4.7m SP02=0.2m SP03=0.2m SP04=11.9m SP05=11.7m SP06=12.0m SP07=11.7m SP08=11.7m SP09=11.7m SP10=11.7m
SP11=11.0m SP12=11.7m SP13=11.0m SP14=11.7m SP15=9.1m SP16=1.9m SP17=1.9m

NOTES

- Pipe/fittings/flanges per class B1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 22.5 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- TV-1045 station on platform at E-113 (EL 103.0) - access by stair from grade.
- Riser at C-101 offset 0.95 m east of nozzle to clear BPA draw and P-112 discharge crossings.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 8"-P-100-053-B1-H BPA RETURN E-113 TO C-101		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DRAWING NUMBER CFU-100-PI-ISO-008	DATE 2026-10-02	REV A

LINE NUMBER 8"-P-100-053-B1-H	FROM E-113	TO C-101 tray 23	PIPING CLASS B1 - 300# CS ASTM A106 Gr.B
DESIGN P / T 13.5 barg / 290 C	OPER. P / T 3.69 barg / 223.9 C	TEST P (HYDRO) 22.5 barg (1.5xPxST/5=1.11)	PIPE WALL SCH 40 (8.18 mm)
INSULATION H 75 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 2 - T > 150 C or equipment-connected
P&ID CFU-100-PR-PID-003, CFU-100-PR-PID-007	FLUID / PHASE P / L BPA return to column	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 17 spools; 38 shop / 15 field